1.

Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

ANS:-

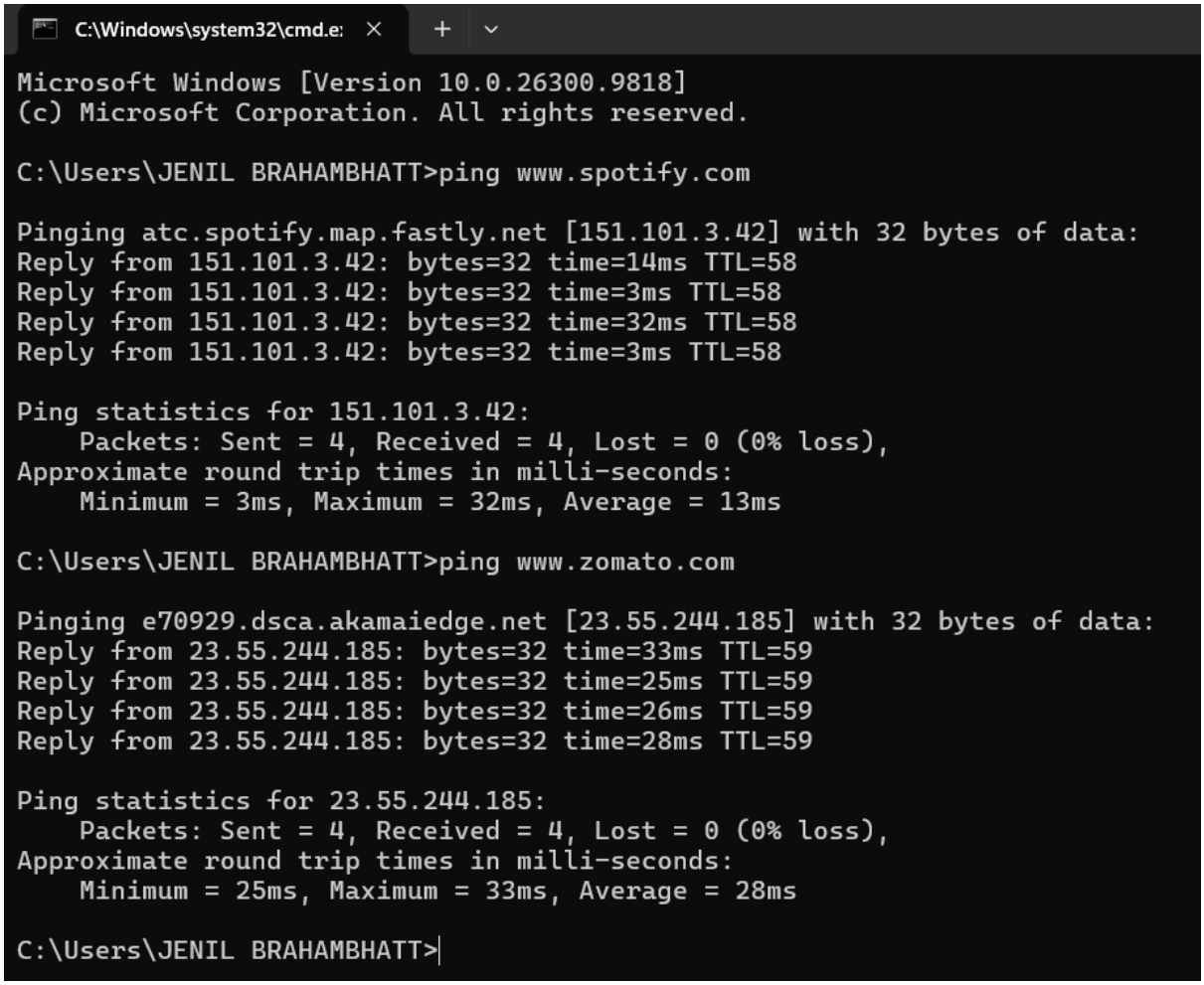
Top 3 Protocols Observed (via *Statistics > Protocol Hierarchy*):

1. TLSv1.2 / TLSv1.3 (Transport Layer Security): Accounts for the highest percentage of traffic, as modern web platforms and media streaming services encrypt all user data and video packets.
2. TCP (Transmission Control Protocol): Operates at the transport layer to guarantee reliable, ordered, and connection-oriented packet delivery beneath the encrypted sessions.
3. DNS (Domain Name System): Handles queries and responses translating domain names (such as youtube.com and instagram.com) into numerical IP addresses during navigation.

2.

Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.

ANS:-



- www.spotify.com:
    - Individual response times: 14ms, 3ms, 32ms, 3ms
    - Average Round-Trip Time: 13 ms
- www.zomato.com:
    - Individual response times: 33ms, 25ms, 26ms, 28ms
    - Average Round-Trip Time: 28 ms

Conclusion

www.spotify.com responds faster with an average latency of 13 ms (compared to 28 ms for Zomato).

3.

Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.

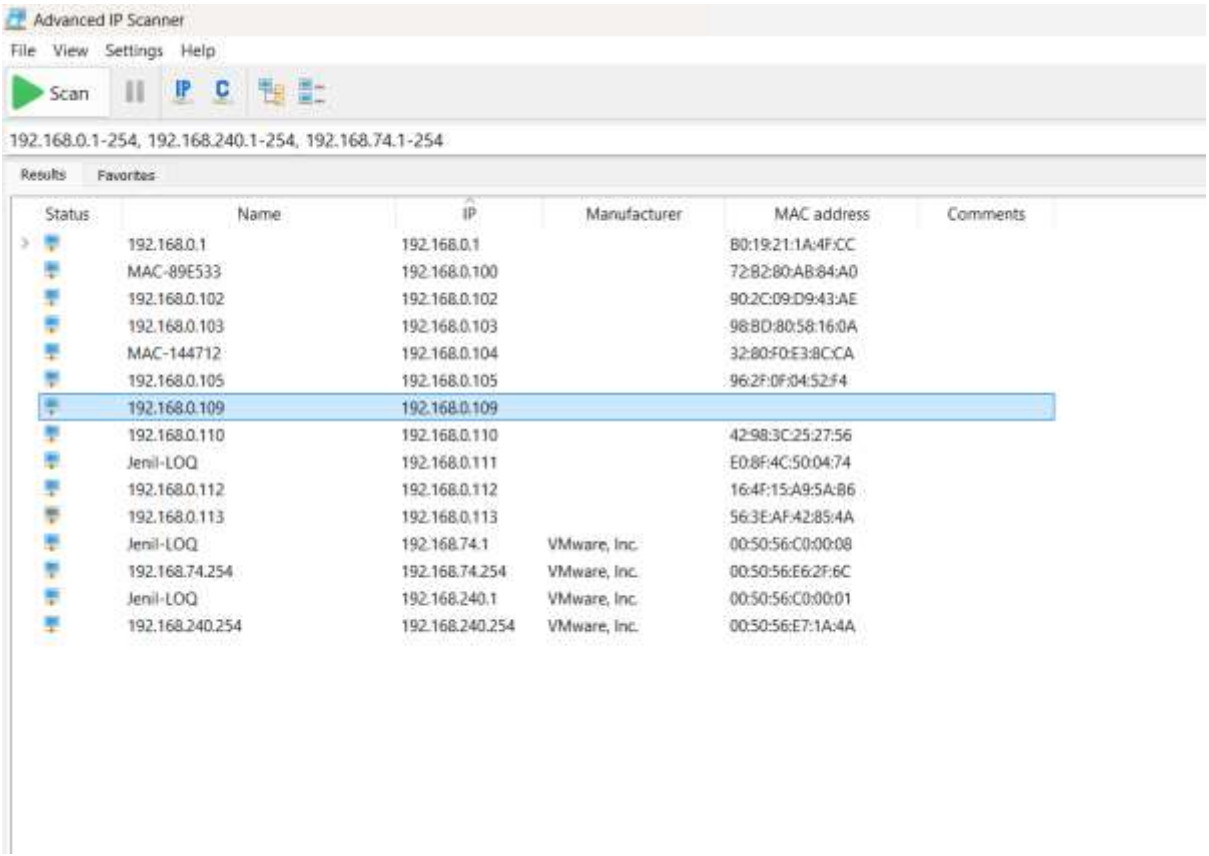
ANS:-

Application / Process Name	Description	Typical Memory Usage
Google Chrome (chrome.exe)	Main browser process and GPU/utility handlers	~350 MB – 600 MB (combined across tabs)
WhatsApp (WhatsApp.exe)	Electron-based desktop chat application	~180 MB – 300 MB
Microsoft Edge (msedge.exe)	Alternative web browser instance	~250 MB – 450 MB

4.

Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network. Hint: Make sure your firewall allows scanning, and only scan your own home/college network.

ANS:-



Device 1:

- IP Address: 192.168.0.111
- Device Name: Jenil-LOQ (MAC: E0:8F:4C:50:04:74)

Device 2:

- IP Address: 192.168.0.100
- Device Name: MAC-89E533 (MAC: 72:B2:80:AB:84:A0)

5.

Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

ANS:-

1. **Network Cable Tester:** Used to verify continuity, correct pinout wiring, and detect breaks or miswirings in an Ethernet (RJ45) patch cable.

2. **Crimping Tool:** Used to attach RJ45 connectors securely onto the ends of raw Cat5e/Cat6 UTP Ethernet cables when making custom cable lengths.
3. **Digital Multimeter:** Used to check electrical continuity, measure voltage levels, and troubleshoot power supply issues on network hardware or power adapters.